CellForge: From Authored Numerical Cells to Competing Offline Circuits

Mechanism evidence, local cooperation, and the unresolved transfer problem

CellForge project*

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Abstract

Can a language model transfer a small, explicit computation into numerical weights, and can those cells improve through competition and cooperate without the author remaining in the inference loop? CellForge studies this question with literal cell and router tensors, counterexample-informed authoring, numerical inheritance and separate cell and circuit admission. We distinguish three claims: executable authorship, useful cooperation, and sustained improvement on untouched tasks. The current evidence supports the first two in bounded settings, not the third. An authored router solves 318/400 generated grid cases versus 107/400 with uniform routing; numerical crossover children fall to 226/400 and 231/400. Real grid revisions improve exposed development coverage from 0/4 to 2/4, but remain 0/4 on each reserved reporting batch. Independent grid-cell competition admits no new children over two generations and solves 0/8 reporting tasks. Beyond grids, a twelve-parameter arithmetic cell solves 32/32 typed probes, tying a fixed historical baseline. A corrected action-cell population records local argument and router repair selection, but its reported 4/5 fixture total is invalid as benchmark accuracy because one target contradicts the request. The subsequent quality verifier withholds runtime and scientific acceptance. A separate comparator/multiplexer prototype selects whole topologies and scores 8/8 reused finite-domain cases, with random controls also perfect in two contexts; its runtime verification gate remains unmet. These findings locate the unresolved problem: valid numerical modules and useful routing do not yet yield a growing, generalizing population. We specify matched-budget ablations and an independent cell-to-circuit test that could falsify the proposed route from authored modules to reusable offline computation.

1 Introduction

A small computation can be represented in several ways: as a program, as fitted neural weights, or as weights written explicitly by an author who understands its mechanism. The third possibility creates a concrete research question. If a language model can write useful numerical cells from examples and revise them from failures, can selection retain those cells, numerical inheritance reuse them, and learned routing combine their competence into an offline circuit? The artifact, rather than the authoring LLM, must perform the final computation.

CellForge's hypothesis is that *numerical cells can be the unit of both variation and cooperation.* A cell first competes with compatible cells on its own behavior. Only cells with measured development competence become eligible for assembly. A circuit or mixture of experts (MoE) then competes as

*Authoring provenance includes Claude and configured Codex GPT-6.1 Sol tasks; run receipts identify the actor and model. Project owner: [name to be supplied].

a separate artifact, with its own numerical router, lineage and admission archive. This two-level hypothesis is stronger than choosing slots from a known library or showing that a hand-assembled ensemble executes. It predicts retained competence, useful cooperation and eventual improvement on untouched task families.

Three claims, three tests. *Authorship* asks whether the saved literal weights control the claimed computation. *Cooperation* asks whether the joint artifact improves over its constituent cells and simple routing controls. *Learning* asks whether a budgeted sequence produces admitted descendants that improve on an untouched reporting distribution without losing earlier skills. A tensor edit, a passing interface test, or a positive demonstration score answers only part of this chain. Calling all three "evolution" hides the distinction that matters most.

Why these experiments matter. Synthetic routing separates a missing selection mechanism from missing expert competence: the router improves while its expert oracle remains fixed. Numerical inheritance then tests whether compatibility preserves that competence; the children lose accuracy. Real grid tasks test whether development-informed structure additions generalize; current reporting is zero. Independent cell competition tests whether an actual lineage grows rather than an ensemble being repeatedly rewritten; the grid lineage stalls. Arithmetic and file-action niches test whether the same numerical representation can leave the grid domain. They provide positive execution and local-admission evidence, but do not remove the need for independent circuit evolution or broader transfer.

Scope and evidence boundary. The primary track uses no gradient-based fitting, no fitness-weighted gradient estimators and no inference-time LLM. Its maker imports competence from a pretrained LLM, while typed encoders, extraction rules and executors supply additional structural priors. These contributions are reported separately. Public-training ARC tasks reserved from the current maker are not private benchmark evaluations, and pretraining exposure is unknown. Corrected controls can revalidate a mechanism on a frozen batch, but cannot make an already observed reporting batch untouched again. Earlier independently evaluated digit recognizers and synthetic-language models establish feasibility baselines; their results are preserved in Appendix A and are not evidence that the current population loop has succeeded.

Contributions. We provide (1) an executable representation and provenance protocol for authored cell and router tensors; (2) a two-level competition contract that separates individual-cell selection from circuit admission; and (3) receipt-backed positive and negative experiments that identify which links in this proposed construction process work and which remain unproven. The central finding is a gap, not a completed scaling result: bounded useful modules and routing are possible, while sustained admitted learning across tasks has not yet been demonstrated.

2 Related work

Architecture search composes motifs hierarchically [1] and evolves topologies and components [2]; both fit weights by training. Our evidence memory adapts ideas from visual retrieval-augmented generation [3, 4, 5]. Neural cellular automata have been applied to ARC-AGI with a separate gradient-trained model per task [6]; our evolved variant (Section A.9) replaces gradients by discrete selection and is a different experiment. Tracr compiles hand-written programs into transformer weights without gradient descent [7], the closest prior example of directly constructed weights that

we recorded. A recorded report of LLM-seeded genetic search over fixed-width programs on ARC-AGI-2 [8] motivates our validity-preserving genome; we did not independently verify its numbers and do not compare against them. We are not aware of prior work in which LLMs author all weights of a working recognizer without fitting, but a wider prior-art review is required before any novelty claim.

3 System

3.1 Representation

A *cell* computes $h = \sigma(W_1 x + b_1)$, $y = W_2 h + b_2$ over the last axis of its input. A *bundle* is a typed DAG whose nodes are parameter-free operations, parameter tensors, cells, other bundles, or a top-1 `dispatch` among bundles; unselected branches do not execute. The same representation is used at every depth; definitions may be shared and parameter sharing is explicit. Recursive template references and exponential expansion are rejected at validation. A model has an `encode` bundle (pixels $\rightarrow$ memory) and a `step` bundle (memory, step index, previous token, state $\rightarrow$ logits, state), driven by a bounded greedy decoder that stops on the model's own EOS. Structural profiles (ladder, ladder-micro, open) constrain organization and are checked by a validator.

3.2 Construction protocol and maker operator

Makers submit JSON proposals of atomic operations (set parameter by explicit value or by a deterministic expansion of an author-written pattern; put/delete definitions; wrap/unwrap). Proposals are validated against their exact base hash, executed on sample inputs and, for pure refactors, tested for exact behavioural equivalence. Makers may not run code that computes weights from data, may not script search loops, and receive development feedback only through a budgeted `dev-eval` command. A manual-maker operator records development train-pair diagnostics, authored transactions, actor/model provenance, graph hashes and candidate status. Each serializable proposal consumes a budget unit, including invalid, rejected and unchanged edits; transactions are applied to a copy and checked against graph validation, structural profiles and typed runtime fixtures. Three focused tests pass. The operator calls no LLM service and performs no benchmark itself, so a matched non-LLM mutation arm must use the same proposal budget and access. A frozen candidate receives one ledger-accounted final call, consumed even on failure. This is accounting infrastructure, not capability.

3.3 Current literal-weight protocol and structural provenance

The owner's 30 September standard (`docs/ONE_STANDARD.md`) defines a joint genome as literal cell and router tensors, with LLM-written founders and counterexample-informed numerical mutations, charged proposals, matched random controls and numerical inheritance. Earlier fixed-gene socket searches remain baselines; their scores do not answer this mechanism-level question. A complete milestone additionally requires useful held-out routing, real-anchor nonregression, resource and false-positive evidence, and competition within and across niches. A valid tensor transaction or passing implementation test alone is insufficient.

The current joint checkpoint reuses four existing specialist structures and their recorded cell-founder provenance: bounding-box, color-map, convolution and neighbor-count computation. Its 78 cell parameters and 20 router parameters form one artifact. Newly authored literal router tensors select experts using observable demonstration mismatch and query input; query targets and niche

labels are excluded. The expert oracle uses targets and is diagnostic only. The count-to-convolution compatibility map is explicit structural glue: eight corresponding off-center kernel rows and bias inherit donor values, while the recipient center row and paint weights remain. The donor router row is copied as part of the child. This mapping is not evidence that an LLM discovered the runtime architecture; reverse transfer rejects incompatible non-isotropic kernels rather than averaging them.

The real-task continuation expands the palette to ten colors and inherits the four expert computations. Its first revision changes six color-map scalars. Its second revision authors literal tensors for a fifth, alternating-row stencil expert and extends the numerical router to five experts, retaining the inherited four-by-four block and biases. The executor supplies the row-zero input condition, row parity, local stencil traversal and output selection: these newly implemented structural operations must be credited separately from the authored numerical values. The development-informed structure is not blinded discovery. Multiple structures and eventual visual/reasoning domains are goals; these grid experiments do not establish cross-domain inheritance or one successful model across ARC-1/2/3.

3.4 Two-level competition, numerical inheritance and admission

A live cell population is indexed by typed structural/skill niche, stores literal weights and keeps historical opponents after retirement. A maker proposal must name live parents, preserve numeric provenance and pay for every edit, including invalid or neutral proposals and structural/interface repairs. Within a niche, admission requires measured fresh development gain and preservation of archive, anchor, false-positive and resource constraints. Reporting cannot select parents or enter the admission archive. Unknown required evidence fails closed.

An eligible cell can contribute to a typed circuit or MoE; eligibility does not admit that circuit. Whole-artifact competition requires its own lineage, router variation and admission archive. A cell can improve while its circuit regresses, and a useful circuit can contain individually weak or redundant cells, so both levels need independent evaluation. The current grid pilot compares assemblies but lacks completed independent circuit evolution; the action pilot executes cell admission but couples the winners deliberately. A separate logic-circuit prototype now executes topology/glue competition, but its missing independent replay and pre-run gate leave admission provisional. The full two-level gate therefore remains incomplete.

Numerical crossover exchanges compatible parameter groups, with explicit mappings and inherited-versus-authored receipts. Earlier `import_def` and fixed-slot experiments test representation validity and wiring baselines; they are not the primary literal-weight evolutionary claim. Compatibility establishes that a child executes, not that it preserves parental behavior or improves fitness. Cross-domain inheritance requires a measured typed alignment rather than renaming a grid socket.

3.5 Governance and evaluation boundary

A hash-chained ledger, graph version store and receipts make each run auditable; reported ledger-entry counts were verified after retrieval. Final cases for the digit task exclude, for strings of length ≥ 4, every string authors ever saw; the unseen-font split uses a font file never present on the authoring machine. Reserved audit data must never be shown to a maker, and schema guards alone cannot prove information isolation.

4 Experiments

Unless stated otherwise each experiment is a single run per arm.

4.1 Literal cell/router weights: synthetic routing and real-task nulls

Synthetic joint checkpoint. Saved `summary.json`, raw maker responses, frozen inputs and `inheritance-receipts.json` under `runs/weights-pop1-router-v1/` record the following exact whole-case results on 400 generated reporting cases, frozen separately from development. The

Artifact	Actual / 400	Target-aware oracle / 400	Uniform / 400
Cold founder	107	345	107
LLM router mutation	318	345	107
Count-to-convolution child	226	258	107
Within-convolution child	231	260	107

Table 1: Literal joint-weight checkpoint. The oracle is an upper-bound diagnostic, not a deployable router. Both inheritance children lose capability relative to the learned-router parent; no crossover advantage is demonstrated.

cross-structure receipt records 20 changed convolution weights, one bias and one router weight; the within-structure receipt records 24 changed convolution weights, one bias and one router weight. These are executable numerical transfers, not socket selection. Copying a cold donor router row confounds the operator comparison. The learned router remains 27 cases below its expert oracle. Random-router controls score 100, 107 and 101/400; local numerical-mutation controls score 70, 49 and 166/400; replacement controls score 0, 219 and 16/400. The three seed contexts reuse authored proposals and deterministic children, so they are not three independent maker replications. Five proposals per arm per context give 30 charged proposals and 14 distinct genomes. Comparisons are conditional on a frozen learned-router parent, not accepted population trajectories. All candidates score zero on four public ARC anchors and none is admitted; resource-cap calibration is fail-closed. Synthetic routing is demonstrated, whereas ARC transfer and accepted evolution are not.

Historical headline is provisional. The saved first-generation summary in `runs/weights-pop1/summary.json` reports 85.69% LLM mutation, 74.07% random and 73.83% breeding. These aggregates are retained as historical receipts, not validated evidence of superiority. Earlier checks flagged mixed-task nontriviality, recomputed LLM superiority and router presence/usefulness; the current mandatory rerun passes 23/25 tests, with historical nontriviality and missing-router failures remaining. The later joint continuation does not retroactively validate this earlier protocol.

Bounded real-task cycle. The sealed `runs/weights-real-learning-v1/` receipts pin twelve ARC-AGI-2 public-training tasks at source commit `f3283f7...`: four development, four archive and four task-held-out reporting tasks. Reporting tasks are held out from the current maker, not private competition cases; earlier author-model exposure is unknown. Nine joint genomes and eighteen charged arm/seed proposals are recorded. The standard manifest and hash-chained ledger were packaged after execution from frozen records, not recorded as an original online ledger. The coordinator now knows the reporting outcomes; further tuning requires a new untouched reporting partition. Shared maker responses across contexts are not independent author replications. `learning_curve.json`, `admission-results.json`, maker responses and `revision1-receipt`

Artifact	Development / 4	Archive / 4	Reporting / 4
Inherited parent	0	0	0
Numerical revision	0	0	0
New structure plus router	1	0	0
Random controls, each of three seeds	0	0	0

Table 2: Real-task exact solutions from the frozen learning curve. Random controls are zero at both revision generations. The single development gain does not transfer to the four reporting tasks and does not result in promotion.

`.json` distinguish evaluation from admission. The first revision has no numerical task gain; the second solves development task `3ac3eb23`, but required false-positive comparisons contain undefined values, so admission remains closed. No candidate is promoted and no held-out transfer is observed. Local resource receipts measure traced allocations, not process RSS; their passing flags cannot certify the full deployment envelope.

Separate Hetzner continuation (v2). Retrieved `summary.json`, `PREDECLARED.json` and `child-receipt.json` from `weights-real-learning-v2-20261001` record one further structural/numerical mutation, evaluated on the existing server. The child `072d019c...` preserves five expert computations and the five-by-five router block, adds a sixth expert and 35 numerical parameter positions, and is not a crossover child. The maker is Codex GPT-6.1 Sol with low reasoning. Fixed four-connected component extraction and geometric feature predicates remain executor priors, not newly learned weights. The inherited v1 parent solves 1/4 development tasks, the child 2/4, and matched random controls 0/4, 0/4 and 1/4 for seeds 11, 22 and 33. All five candidates solve 0/4 on a new reserved public-training reporting partition (task IDs `0becf7df`, `689c358e`, `60b61512`, `2281f1f4`); their target-aware expert oracles also solve 0/4. Archive and anchor totals remain zero, and no promotion or held-out gain occurs. The new reporting partition is separate from v1 and is not an official private evaluation. One shared authored candidate and three seeded controls are conditional comparisons, not independent maker replications. The saved server run starts at 00:38:18 UTC on 1 October, records 2.043 seconds of evaluation runtime and 39,680 KiB peak RSS; this excludes authoring, transfer and subsequent sealing, so it is not full-cycle time or cost. These receipts demonstrate a further development gain while the primary held-out learning criterion remains unmet.

Verification limits. The real-run semantic receipt passes nine integrity/provenance checks but fails three scientific assertions: useful held-out skill relative to zero weights, recomputed LLM superiority, and useful held-out routing. Compatibility tests, including exact reproduction of donor outputs after directional repair, establish execution semantics only. They do not supply missing task generalization, accepted competition or a crossover advantage.

4.2 Independent cells, non-grid computation and action repair

Cell selection is a separate experiment. The first independent-cell pilot, `runs/weights-cell-population-g1/`, extracts six historical numerical experts into typed niches. Two authored proposals, one aligned crossover and three random controls are evaluated in one seed context; all six proposals are rejected. Two cells earn development-based assembly eligibility. Their deliberately assembled MoE and the historical full parent both solve 2/12 development tasks and 0/8 reporting

tasks. The assembly uses fewer eligible cells but does not establish a new circuit lineage or circuit admission. This pilot is neither three-seed replication nor a balanced three-arm comparison; the standard suite flags its seed count and budget mismatch.

The subsequent `runs/weights-cell-sequential-g2/summary.json` records two sequential generations with four charged proposals per arm per seed context, 24 charges in total, and two shared physical maker responses. Six live founder genotypes remain; no new child is admitted, ancestry depth remains zero, and all 22 evaluated genotypes solve 0/8 reserved reporting tasks. Candidate mean development accuracy rises from 0.0625 to 0.125 in the authored arm, while the random arm remains zero, but this is not an improving live population. A candidate trajectory must not be presented as a champion trajectory when selection rejects every candidate.

A numerical cell outside the grid domain. A twelve-parameter decimal arithmetic cell takes typed digits and carry as input, uses authored linear weights and a threshold to return a digit/carry pair, and solves 12/12 development and 32/32 reporting probes. Three controls randomize the same twelve positions and bounds, each scoring 0/32 with invalid outputs. However, the fixed historical arithmetic artifact also scores 32/32. This tests whether the numerical cell implements a known algorithm, not whether authoring beats a competent scripted baseline or learns language. The input contract, digit bounds, threshold and output validation are supplied by the executor; no natural language parsing or real-benchmark admission occurs. Source: `runs/weights-cell-sequential-g2/arithmetic-summary-controls.json`.

Logic cells and provisional circuit competition. Existing literal threshold-weight cells implement Boolean operations, two-bit comparison, multiplexing and arithmetic. Independent exhaustive truth-table checks pass, including a half-adder/full-adder composition; additional component checks cover bounded combinational functions and finite-state streams. These 11 implementation tests establish exact behavior on their declared finite domains, not learned reasoning or length generalization. Source: `runs/baseline-logic-math-cells/`.

A subsequent server prototype selects *whole circuits* rather than merely coupling one chosen pair. The comparator and multiplexer founders are independently measured, then seven authored topology/glue proposals compete against seven paired random proposals in each of three contexts. Both search arms are charged 27 new numerical positions per context; an unconnected control is charged separately. Development selection requires correct typed execution and a contribution drop when comparator, multiplexer or glue weights are ablated. Each circuit archive records its selected artifact and admission events separately from cell records. The task computes the maximum of two two-bit numbers over the complete 16-input domain, split into eight development and eight reused reporting cases.

Saved `SUMMARY.json` records one selected authored circuit in all three contexts, scoring 8/8 reused reporting cases. Random circuits score 6/8, 8/8 and 8/8; two random contexts also record a circuit admission. This demonstrates executed topology/glue competition in a small known domain, not LLM superiority or fresh generalization. The reporting split was already used in a worker trial, and requested maker metadata does not independently observe backend identity. Crucially, the sealed `VERIFICATION.json` withholds both runtime and scientific acceptance: independent circuit replay was not implemented and the pre-run circuit gate was not executed. Thus archive "admission" events are provisional prototype selections, not verified runtime admission. The source receipts reside in the retrieved `logic-circuit-population-results.tar.gz` from `/srv/cellforge-inbox/weights-logic-circuit-population-g1-20261001/`. This is a concrete advance over fixed deliberate wiring, with an explicit remaining verification gap.

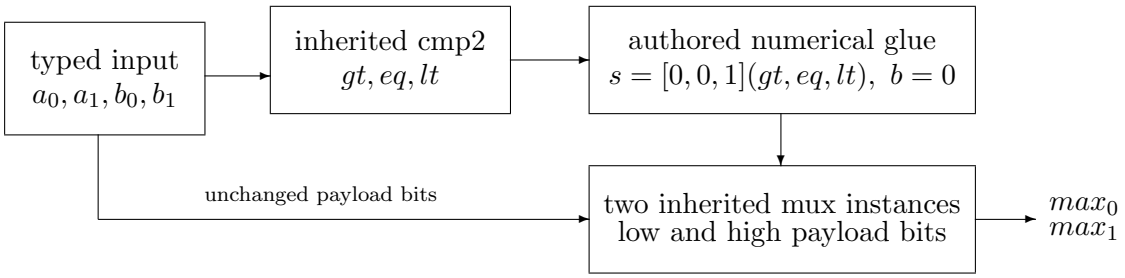


Figure 1: Actual development-selected `lt-selector` topology from the logic circuit prototype, not a proposed future architecture. The comparator and multiplexer contribute 46 unique inherited numerical positions; two executions of the shared multiplexer give 57 expanded inherited positions. Only four glue positions are newly authored in this candidate. Its 8/8 reused reporting score also occurs in two random-control contexts. The circuit's runtime verification is incomplete, so selection is provisional.

Synthetic tool abstractions are not tool execution. The separate `runs/weights-tool-router-v1/summary.json` reports authored joint routed accuracies 0.930, 0.903 and 1.000, versus 0.250 with uniform routing, in three seed contexts on generated tool-cell abstractions. The maker also wrote the generators and knows their rules; niche-normalized scores differ from raw mixed-task scores. Its 36 crossover children include 13 that exceed the better parent and 20 that are worse, but crossover is uncharged and applied to both lineages. This is not a matched crossover-advantage experiment and it does not execute real tools. It motivates the stricter action experiment below.

Corrected local action-population mechanism. In `runs/weights-action-population-corrected-g3/`, literal numerical argument cells select copied spans, a separate numerical router chooses the operation, and a typed adapter builds the action packet. The executor receives that packet rather than inferring intent or filling missing arguments. Exact resulting files, not just action labels, determine fixture success. Argument and router cells compete independently against recorded historical opponents. Generations one and two reject their authored argument repairs; generation three admits the argument repair, and the separate router repair is admitted locally. Each context records two admissions and two retirements, leaving one live cell in each niche. These selected cells are deliberately coupled through the adapter; no evolved circuit topology or separate circuit admission is claimed.

Control correction matters. The earlier router control changed 261 positions, whereas the author could change 128. The corrected run gives both the same 128 router positions, and matches argument masks of 27, 27 and 30 positions across three rounds, with bounds $[-8, 8]$, $[-8, 8]$ and $[-9, 9]$. Each arm receives three argument proposals and one router proposal per context. Authored responses are shared across seeds 101, 202 and 303; these are three random-control contexts, not three independent maker replications. Previous admissions are quarantined and selection replayed from the founders.

The raw receipt records 4/5 reporting fixtures in each context, but this is an invalid five-case benchmark metric, not a valid accuracy estimate. The batch is *unchanged correction validation*: aggregate results were already exposed, and some grammar was visible during authoring. It is not fresh held-out evidence. There is also a target inconsistency: fixture `newbare` requests a write to `LOG` but the manifest expects `JOURNAL`; the executor writes `LOG` and the receipt counts that case wrong. We preserve the recorded 4/5 rather than silently changing the target or crediting

an additional success. The target contradicts the requested file effect, so the five-case benchmark metric is invalid. The subsequent `QUALITY_VERIFICATION.json` fails closed with `fixture_file name_not_copyable:reporting:newbare:JOURNAL`. The original receipt is preserved as historical raw output; it must not be cited as valid 80% accuracy. A new, consistent frozen corpus is required before any capability comparison.

Preventing a host-side bypass. The source-bound runtime gate executes tests that vary weights, check resulting packets and file effects, reject missing arguments and escaping paths, and verify that the adapter does not recover hidden intent from the request. Independent replay binds raw proposals and source hashes, recomputes predictions and reconstructs budgets, live-parent selection, births, retirements and historical opponents. The earlier `RECOMPUTED_VERIFICATION.json` recorded runtime acceptance. The later `QUALITY_VERIFICATION.json` supersedes that permission: evidence quality and mechanism replay fail, runtime acceptance is false, and scientific acceptance and headline permission remain false. Population replay still reconstructs local selection, budgets and memberships. Attribution tests show that numerical cells affect action outputs, but do not validate the defective benchmark. No language understanding, general-purpose agency, ARC transfer or completion of the two-level architecture is established.

Measured execution costs, with their boundary. The sequential grid-cell receipt compares first-generation scoring of eight unique candidates on two workers (0.311 s) with serial verification (0.571 s), approximately 1.84× for that scoring batch. Generations remain sequential, scores agree, and shared/cached candidates do not create new author replications. The whole first server stage takes 1.071 s; generation two's scoring takes 0.337 s, without a paired serial measurement. Transfer receipts distinguish a 102.521 s generation-one feedback download, a 3.155 s compressed generation-two upload and a 7.589 s final compressed receipt download. These are different phases, so their ratios are not transfer speedups. Neither the scoring comparison nor these network timings measure end-to-end training acceleration or total authoring cost.

Test	Observed result	What the result establishes
Generated joint grids	318/400 routed; 107/400 uniform	Useful bounded routing; inheritance children regress
Real grid continuation	Up to 2/4 development; 0/4 new reporting	Development repair without measured transfer
Independent grid cells	0 admitted descendants; 0/8 reporting	Executed competition with stalled lineage
Typed arithmetic	32/32; random 0/32; fixed baseline 32/32	Known numerical algorithm, not learned superiority
Logic circuit prototype	8/8 reused; random 6/8, 8/8, 8/8	Topology competition executes; runtime gate missing
Corrected action coupling	Raw 4/5 is invalid benchmark accuracy	Local selection recorded; quality gate fails; no runtime/scientific acceptance

Table 3: Current evidence by claim. Denominators refer to different corpora and must not be pooled. The historical action total comes from a previously observed batch with a contradictory target and is invalid as five-case benchmark accuracy.

5 Discussion

The result is a partially completed causal chain. The strongest current positive evidence is that authored numerical values control bounded computations and that routing can turn available specialist competence into more correct joint predictions. The decisive negatives are inheritance regression, zero real reporting solutions and a stalled independent grid lineage. The action population records local selection and measurable state changes, but its inconsistent fixture invalidates the accuracy headline and closes the quality gate. Deliberate action coupling therefore supplies no accepted transfer result. The separate logic prototype executes circuit variation and selection, but lacks its independent verification gate and shows no advantage over two random controls. Together these results justify testing the construction mechanism further, not declaring that scaling has been achieved.

Competing explanations require explicit controls. A gain may come from the author's prior knowledge, a new executor primitive, extra structural capacity, an easier generator, additional feedback or a larger edit mask. Uniform routing isolates one cooperation question but is weaker than best-single-cell and development-selected fixed routing. Random weights test the value of numerical authoring but are weaker than a competent scripted algorithm. The arithmetic tie and corrected action budget illustrate why both kinds of baseline are necessary. No current result isolates a general crossover advantage or the incremental value of a separately evolved circuit population.

Research-assistant goal drift: substituting an easier proxy. The original objective was offline LLM authoring and evolution of numerical cell weights. Early receipts contain literal `set_param` proposals: nine operations in `runs/p1/tasks/r1-iter-a/proposal.json` and 79 in `runs/lm1/tasks/r1-L1/proposal.json`. Later grid experiments instead recombined socket assignments among fixed genes (`runs/arc-grid-genes/expanded-breeding-v2`). This observable mismatch between the requested mechanism and the delivered proxy is a research-assistant workflow failure, not evidence that the research objective changed. Crossover tying mutation on those socket genotypes cannot establish whether actual weight evolution is superior. The owner's humorous description that the assistant "got lazy" is an interpretation, not a measured motivation or causal explanation. These artifacts establish the substitution, but do not identify its internal cause; a mechanism-level acceptance check should require recorded numerical changes and compatible lineage before treating future runs as tests of weight evolution.

Current mechanism and remaining gap. The literal-weight continuation repairs the earlier proxy mismatch by recording numerical cell/router edits and compatible inheritance. Its synthetic router gain is evidence for routing in this four-color setting. The real-task null and failed admission gates leave the main objective unfinished; neither interface tests nor more authored structures establish independent ARC learning.

What worked. Direct authoring produced working recognizers and compact language machinery without gradients: digit strings read perfectly in distribution, with one artifact retaining 73.8% under an unseen font, and authored synthetic-grammar models reached within 0.004 bpc of the generator entropy. Typed crossover retained parental building blocks on tiny probes and lifted MNIST-1D accuracy by 0.064 in one selected child. Narrow authored components (typed transfer, binding, deduction, addition, action grammar) execute exactly within their grammars, and a compiled backend gives large verified speedups.

What did not. Authored and control character models sit at chance on public multiple-choice probes; the byte language graph emits invalid actions and repeats tokens; document evaluation is blocked on language capability and on an unresolved related-edition overlap; evolution from authored founders tied random search in the language mini-pilot and in most ARC pilots; dense synthetic success (300/300) did not transfer to DEV200; subsequent targeted additions reach only 5/200; and evolved NCAs solve 0/40. The gradient MLP beat the best MNIST-1D authored model.

Honest limits. Final evaluations exist only for the digit, synthetic-grammar and MNIST-1D experiments; all ARC, byte-language and document results are development results on exposed public data. Most experiments remain single small runs; the nine-slot study has three predeclared replications and favors random search. Small earlier gaps do not establish general superiority. Authors and orchestrator share an operating-system account locally, and the orchestrator has administrative access to the evaluation server, so the final-evaluation boundary is physical rather than adversarially audited. Token usage is self-reported; the digit task used a single font family in development; the visual-retrieval memory arm and the MNIST-1D harness-version comparison were never run; the MNIST-1D crossover has one independently tested child and no random-crossover control, so its advantage may reflect selection and extra computation; a co-researcher's library results and scratch NCA numbers are not reproduced by us. The authored models transfer competence from the author LLM's pretraining and must be reported as such. There is no private-set score, no official ARC-AGI-2 or BFCL score, the system is not an LLM, and no claim of general intelligence or frontier superiority is made.

Design-only pretrained composition. A System 1/System 2 proposal (`docs/SYSTEM12_DESIGN.md`) would compose pretrained models (a gradient-trained 0.6B-parameter decision-head system and a pretrained generative candidate) with evolved routing. It would import pretrained capability, is separate from the no-gradient line, and was not executed: no model was downloaded, and the approximately 4.34GB required files exceed the reported 3.5GB free disk.

6 Next falsifiable tests

The next decisive test is a predeclared two-level experiment, rather than another known-rule fixture. Freeze several typed skill families, task-level development and reporting partitions, numerical/structural edit vocabularies and total charged budgets before authoring. Run independent maker replications; random seeds alone are not independent maker trials. Compare counterexample-informed authoring, matched random edits and competent scripted repairs, then numerical crossover against a matched no-crossover arm. Charge invalid edits, neutral proposals, interface repairs and crossover itself. Report both physical unique evaluations and budget charges.

First require admitted cell descendants to beat founders on fresh development cases while retaining historical opponents and real anchors. Then freeze the live cells and compare the best single cell, uniform routing, development-selected fixed routing and an authored numerical router. Only afterwards test circuit evolution with its own live-parent lineage and admission archive. A positive result requires untouched reporting gain, retained earlier competence and an advantage beyond extra parameters, feedback or evaluation budget. Reject the proposed mechanism at the tested scale if lineages repeatedly stall, mixtures add no independent competence, or crossover losses dominate matched controls.

The action track needs a new frozen corpus with internally consistent requested and expected file states, held-out grammar combinations and adversarial abstention cases. Attribution ablations

must disable the numerical router, numerical argument selection and structural feature extraction separately; the executor must still receive only a typed packet. The arithmetic track needs harder composition and length tests against its already-perfect fixed baseline. OCR/reasoning distillation is planned or prepared, not a completed experiment. Credit-dependent remote work awaiting credit validity is not a training result. Neither plan supplies evidence for a conversational system or one successful model spanning ARC-1/2/3.

7 Reproducibility

The 1 October two-level mandatory suite runs 47 tests: 43 pass and four fail. The current independent-cell pilot fails balanced budget and three-seed requirements; semantic checks also fail useful nontrivial held-out skill and the required router interface on that pilot. Passing mechanism checks therefore coexist with an unfinished scientific milestone. The corrected action quality replay withholds runtime and scientific acceptance after detecting its contradictory fixture target. Population-selection replay remains valid as scoped accounting evidence. Corpus, source and proposal hashes must be retained together; updating a self-declared result hash alone is insufficient evidence.

Each result names its receipt under `runs/`, with graph and source hashes recorded in the hash-chained ledger; ledger counts were verified after retrieval. Fonts are not shipped (`CELLFORGE_FONT_DIR`), the final-evaluation server holds the private grammar and secret seeds, and the document PDFs are pinned but not redistributed. The pinned document source has 1,082 annotated questions while the README states 1,091; we report the pinned count and have not resolved the difference. The manuscript is built with the existing `scripts/build_paper.sh` entrypoint, using `latexmk` and TeX Live on the evaluation server and fetching the PDF and log locally. On Windows the existing script runs from a persistent Git Bash shell; double-clicking it may close the terminal after completion. The 1 October source update also passes static environment, reference and citation checks. The `paper/check_numbers.py` script lists numeric tokens of this file that are absent from the original draft and the research notes.

A Historical capability baselines and fixed-gene experiments

These experiments retain their original protocols and exposure caveats. They motivate the present mechanism, but do not satisfy the later literal-weight and two-level population standard. Fixed-gene crossover, numerical authoring, fitted controls and independent evaluation must remain distinct.

A.1 Digit strings

Protocol. 32×128 grayscale, 1–8 digits, DejaVu Sans Mono 20 px, horizontal offset 2–14 px, vertical jitter ±2 px, Gaussian noise $\sigma \leq 0.03$. Final splits: iid ($n = 2000$), unseen font Liberation Mono ($n = 1000$), size 18–22 px ($n = 1000$), length 9 ($n = 500$). From an EOS-constant bootstrap: two independent iterative makers with recursive bundles (25 dev-eval budget each), a flat-graph maker, and a one-shot maker (validation only, no metrics). Controls: a gradient-trained slot-MLP (Adam, 400 or 20,000 training images) and (1+1) random direct-edit hill climbing (25 or 20,000 evaluations), all scored through the same reference evaluator.

Results. Table 4. All four authored artifacts are perfect in distribution, using two or fewer development evaluations each, so the in-distribution task is saturated and does not discriminate arms.

Artifact	iid	unseen font	size shift	length 9
Authored, iterative A	1.000	**0.738**	**0.723**	1.000
Authored, iterative B	1.000	0.382	0.379	0.000
Authored, one-shot	1.000	0.427	0.453	1.000
Authored, flat	1.000	0.407	0.570	1.000
Gradient slot-MLP, 400	0.057	0.054	0.050	0.000
Gradient slot-MLP, 20k	0.838	0.100	0.419	0.000*
Random edits, 20k evals	0.000	0.000	0.000	0.000

Table 4: Digit strings: exact-match rate on independent final cases (fresh seeds). Wilson 95% intervals are ± 0.03 or narrower at these n; CER in the project log. *The slot-MLP has eight character slots by construction.

Every maker converged on the same mechanism family (vertical alignment, left-edge detection, a fixed 12 px character pitch, hand-typed 15×12 glyph templates), yet small decisions produced large robustness differences: artifact A's normalized agreement score and ± 3 px search tolerance generalized best; artifact B hard-coded eight character cells and failed every 9-digit string. LLM authors can construct compact recognizers (about two thousand parameters for artifact A) without gradient fitting, and the one-shot author needed no feedback. The results do *not* show a difference between arms (one run per arm), robustness to real-world text, or superiority over gradient training: the control lacks translation invariance and a gradient-trained convolutional network would likely be perfect in distribution. The templates came from *looking at* development images; we regard the author's visual perception as the channel through which learning happens and report it as such.

A.2 Synthetic-grammar hierarchical mixture-of-experts language models (`tiny lm/1`)

The owner judged digit recognition too easy a test of the intended cell→cluster→expert design, so authors built character-level language models with routing and recurrent state doing real work.

Task and arms. Text comes from a private English-like grammar (12 nouns, 12 verbs, agreement that must survive relative clauses, adjectives, prepositions, adverbs, ", and" coordination) held only on the evaluation server. Authors see 2,000 sampled documents and 300 development documents, never the grammar. Vocabulary: 32 tokens; the model receives the last eight characters, the position and its own state. Scoring is bits per character (bpc) under teacher forcing on fresh documents, plus grammaticality of 200 free-running samples (temperature 0.8) judged by the private recognizer. Three authors used an enforced ladder profile (cells inside clusters, clusters inside experts, a router driving lazy top-1 dispatch among at least two experts), two an open profile. Controls fitted from data on the same 2,000 documents: uniform, n-gram counts and a gradient-trained MLP over the same eight-character window (Adam, 74k parameters).

Results. Table 5. All five authored models reach 0.678–0.681 bpc, within 0.004 bpc (under 1%) of the generator's entropy, with 94–99.7% grammatical generations, against 0.824 bpc and 20% for the gradient MLP. The authored models keep singular/plural state across nested clauses in a stack held in recurrent state; the fixed-window MLP cannot. The best model (L2) has 60 routed experts, one per grammatical situation, and only 3 of 60 sampled sentences violate the grammar. The task is exhausted: every author read roughly 40 documents, wrote the closed grammar down, hand-tallied rates and compiled it into tables and state machines. This shows the machinery (state, recursion

Model	bpc	top-1	gramm.	gramm.+novel	experts	params
Ladder L1	0.6796	.810	.966	.788	6	278k
Ladder L2	**0.6776**	.809	**.997**	.884	60	1.69M
Ladder L3	0.6790	.809	.940	.804	4	898k
Open O1	0.6797	.809	.955	.839	–	1.31M
Open O2	0.6808	.809	.939	.775	–	1.56M
Gradient MLP (window 8)	0.824	.780	.201	.135	–	74k
Trigram counts	1.383	.617	0	0	–	66k
Uniform	5.000	0	0	0	–	32
Generator entropy: ≤ 0.6734 bpc (summed choice bits, upper bound)						

Table 5: Synthetic-grammar language models: independent final evaluation on 1,000 fresh documents (bootstrap 95% intervals on bpc about ± 0.002). "Gramm." is the fraction of complete generated sentences accepted by the private grammar; "+novel" also requires the sentence to be absent from the authors' training text.

tracking, dispatch among tens of experts) works end to end at 0.3–1.7M authored parameters. It does *not* show that the ladder helps: an open, unrouted model (O1) matched the ladder with three feedback runs against five to eleven, and ladder authors were required to use routing the task does not need. Nor does it show that authoring scales to data whose generator cannot be read off a few dozen samples. Author-reported cost: 176–231k tokens per run.

A.3 Real-text mixture of experts, delegation and failure-informed mutation

The next step moved to real text, where no closed grammar exists. In a delegated real-text model (hash `82ebe5e5...`), three isolated mutations reused settings from earlier failed proposals. The first worsened development bpc from 2.4748 to 2.4826. A trigram-donor change scored 2.4746 bpc, effectively tied with its parent; an after-period-space specialist scored 2.4743 bpc. These comparisons used the 165-document development set. The specialist was accepted on a separate eight-expert branch; its reference-anchor bpc was 2.46152 versus 2.46236 for the seven-expert parent over 155,239 tokens, with identical 0.5220 top-1 accuracy. Both checks are development feedback and establish no final or frontier-benchmark gain; the main branch remains seven-expert.

A UTF-8 byte lift of the eight-expert text MoE retains authored weights and 225-dimensional recurrent state in one graph with a shared 259-token head. Its inherited experts still use only eight projected bytes. A 387-token ASCII compatibility probe preserves 208 correct predictions, with loss changing from 2.4798989 bits per character to 2.4803443 bits per byte. This establishes input plumbing, not language competence; three ordinary or Unicode prompts repeatedly emit "too" to the token limit (receipts: `runs/unified-byte`). A directly authored repetition-memory mutation reading a 64-token context has parent loss 2.480344 versus 2.512589 bits per byte on the same 387 bytes; both predict 208 tokens correctly with zero scoring failures, and both answer 0/4 author-visible novel key/value binding prompts exactly. The mutation was rejected for promotion (`runs/parallel-reasoning/DECISION.json`); this small development probe is not a held-out reasoning evaluation.

A.4 MNIST-1D direct authoring and crossover

MNIST-1D provides a procedural ten-class task on 40-sample signals. We used 4,000 development-training signals, a separate 1,000-signal development evaluation set and a 1,000-signal reference an-

chor; independent final evaluation used 2,000 newly generated signals per artifact with an operator-side secret seed. Makers were told not to inspect the generator source, a restriction that was procedural rather than technically enforced.

Round A and controls. Three independent makers started from the same zero-logit bootstrap under harness version 1, wrote numerical filters and classifier wiring without gradient fitting, and used respectively four, five and four development evaluations. Table 6 shows the strongest authored parent exceeding the logistic control but trailing both MLP controls (trained on the same split). Class-specific strengths differ substantially; three runs are insufficient to estimate a typical outcome.

Artifact	Final accuracy	Wilson 95% interval
Authored A1	0.4655	[0.4437, 0.4874]
Authored A2	0.4360	[0.4144, 0.4578]
Authored A3	0.3410	[0.3205, 0.3621]
Authored two-parent crossover	0.5295	[0.5076, 0.5513]
Authored three-parent crossover	0.5360	[0.5141, 0.5578]
Gradient logistic	0.3355	[0.3151, 0.3565]
Gradient MLP, 100 hidden units	0.5940	[0.5723, 0.6153]
Gradient MLP, 500 hidden units	0.5995	[0.5779, 0.6208]

Table 6: MNIST-1D independent final accuracy; $n = 2000$ per artifact. The MLP controls were not tuned to match published MNIST-1D baselines.

Crossover. A fourth maker imported the independently authored A1 and A2 filter banks into a child graph and authored a combining rule. After five development evaluations the accepted child scored 0.5295 on fresh final signals, an absolute gain of 0.064 over the better parent; it uses about 29.7 million multiply–accumulates over the final set versus 13.3 million for A1 and 16.4 million for A2. This is one selected offspring, not evidence of a general evolutionary advantage. A three-parent extension scored 0.572 on the development set and 0.557 on the reference anchor ($n = 1000$ each), then 0.5360 on a separate fresh-seed final set ($n = 2000$). Its apparent 0.0065 gain over the two-parent child is not reliable: the final evaluations used different seeds, the intervals overlap substantially, and the third parent raises final-set computation from 29.7 to 39.3 million multiply–accumulates. A version-2 diagnostic plugin reporting class confusion and wrong-prediction margins was validated and activated, but fresh-maker Round B under that harness was never evaluated, so no harness-effect claim is possible. New MNIST-1D jobs were paused under the later document-benchmark priority.

A.5 Frontier-benchmark probes at chance

We adapted character-level models to multiple-choice scoring (`mc-loglik/1`) by summing next-character negative log likelihood over each answer choice, also reporting length-normalized scoring. The vocabulary lowercases text and omits many capitals, digits and symbols, so coverage is imperfect. MMLU test, ARC-Challenge test and HellaSwag validation were staged on the private evaluation server; GPQA was unavailable because access requires dataset authorization.

The first sweep covers a synthetic-grammar authored model and three real-text controls (unigram, trigram, small gradient MLP), each on a fixed 1,000-item subset per benchmark. Length-normalized accuracies span 0.243–0.266 on MMLU, 0.224–0.248 on ARC-Challenge

and 0.230–0.264 on HellaSwag. The delegated real-text MoE (`82ebe5e5...`) on 1,000 items each scores 0.241 MMLU, 0.243 ARC-Challenge and 0.241 HellaSwag (Wilson intervals $[0.2155, 0.2685]$, $[0.2174, 0.2705]$, $[0.2155, 0.2685]$; character coverage 0.9147, 0.9624, 0.9889; receipt `runs/bench_results/server-aggregate-20260930.jsonl`). All values sit around the four-choice chance rate of 0.25 and provide no evidence of frontier benchmark competence; they are negative transfer evidence and a test of the scoring interface, not a comparison to frontier language models.

A.6 Byte-level language graph, component specialists and the document hold

Tool-loop probes. The actual byte-language graph (hash `f0b43dbe...`) was asked to generate tool-loop actions on three authored synthetic prompts (one policy step, 48 generated tokens per action). It produced three invalid JSON actions and zero tool calls or responses (`runs/parallel-tools/server-policy-receipt.json`). This language-policy failure is retained alongside the following component results, each a narrow authored algorithm evaluated by the same research team and not evidence of language understanding:

- Typed finite-state graph (`7f810c68...`), `typed-observation-transfer/1`, 128 fresh numeric payloads of width seven (seed 90218, 640 policy evaluations): transfers 128/128 observations versus 0/128 with observations removed, abstains on 128/128 no-request and 128/128 invalid-observation cases and calls on 128/128 missing-observation cases (`runs/parallel-tools/server-typed-receipt.json`).
- Exact binding graph (`e36653df...`, zero parameters): 128/128 server-generated lexical-variation cases (Unicode, missing keys, prefix distractors, duplicates) match, versus 0/128 for a disabled "unavailable" component (a dependency ablation, not a baseline); receipts in `runs/parallel-reasoning/binding`.
- Carry-addition graph (`43c2a56a...`): 63/63 development additions versus 2/63 for an identity control, all five invalid inputs rejected (`runs/parallel-arithmetic/REPORT.json`); its empty source-hash field limits provenance.
- Frozen digit OCR graph (`54e3934a...`) returns correct text for five authored digit strings; the negative control "hello" yields "50115" and blank input yields empty text (`runs/parallel-vision/server-smoke.json`).
- Positive-Horn deduction graph (`3e9019e4...`): 120/120 development cases (71 entailed, 49 not entailed) versus 56/120 with rules removed, 120/120 under rule-order permutation (`runs/parallel-deduction/REPORT.json`).

Composition. The action-grammar graph `8868695d...` answers two fresh synthetic digit images ("28", "39") correctly through the actual specialist harness, each with one OCR call and zero invalid actions; with one observation withheld it returns no answer and repeats the call twice, and with the tool disabled it returns no answer with zero accepted calls and two invalid actions (`runs/action-language/harness-development-result.json`). The composed graph `e04777f4...` preserves inherited MoE dispatch and gives the grammar graph-owned priority; it is bit-identical to its parent on one ordinary-input test, and generates OCR text "587" under preservation probes (four focused tests), and succeeds on two further digit images ("88", "86"). A matched 387-byte probe gives parent and composed graphs identical loss (2.480344 bits per byte) and 208 correct tokens. The source-stability flag is false because a batch method was appended during execution; the coordinator reports identical serial-method ASTs and the executed snapshot is retained in `runs/action-language/composed-runtime-snapshot`. A tiled variant `ab3cee4e...` preserving

unused inherited arrays passes the strict micro profile with identical 387-byte scores; it retains 81,432 parameter slots, 3,868 nonzero values, 96 cells, 26 bundles, 613 instances and one dispatch, edges rise from 14,327 to 15,618, and a single local run takes 12.097 s versus 9.746 s untiled (about 24% overhead, uncertain). The controlled-English deduction composition `bef61dc6...` calls component `3e9019e4...` in an immutable five-condition harness (`runs/action-deduction/harness-snapshot-v1/result.json`): positive, missing-fact and missing-rule cases answer correctly (3/3), each with one call, zero invalid actions and zero specialist errors; withheld observations give no answer and two repeated calls; disabled tools give no answer, zero accepted calls and two invalid actions. Condition latencies are 163.1, 150.6, 144.1, 140.8 and 107.6 seconds. The grammar is bounded unary controlled English, and repeated unavailable calls and high latency remain failures. The coordinator reports 153 full-suite tests passing in 81 seconds; that is regression evidence, not capability.

Document benchmark hold. The priority benchmark is MMLongBench-Doc (repository); this is a research priority, not a claim that it is the hardest benchmark. All five OCR graph families are digits-only, fixed 32×128 raster interfaces with no question conditioning or layout encoding and cannot yet meet its contract; a digit-only proxy would establish nothing. The pinned source has 135 PDFs and 1,082 annotated questions (the README records 1,091; see Section 7). Private staging completed with zero benchmark evaluations. The first split was rejected after two byte-identical PDF groups and one cross-split conflict (`runs/document-crossover/server-stage-v2.json`); canonical grouping gives 133 unique byte groups, provisionally 107 development and 28 held-out document IDs. The related-edition audit reports 67 candidate pairs, nine crossing the split, and 40/133 groups with inadequate extracted text; a later audit groups 117 heuristic families, which does not verify isolation. The split is not frozen. Alphabet-expanded OCR (`8332b0c6...`) recognizes three of six tiny development strings; its 61 literal-glyph checks test the templates, not accuracy. Three OCR crossover children are frozen without selection. The same-item diagnostic (`document-phase-diagnostic/3`) renders in 1.6596 s, completes all 2,125 OCR crops in 133 batches (1.7213 s), then enters language generation at 2.1396 s with a 783,438-byte context and times out at its 120-second limit (`runs/document-crossover/server-phase-diagnostic-v3.json`; the recorded `language_seconds=0` is an unfinished timer). A compact serialization (v4) cuts the context to 185,055 bytes (a 76.4% reduction) yet still times out after 1.7736 s of OCR (`runs/document-crossover/server-compact-diagnostic-v4.json`); this negative result selects no model. A compiled-runtime backend initially shows about 1.4× step speed; differential qualification (29 tests, 74-fixture injection suite; first NaN failure retained and repaired) reports amortized speedups of about 21–26× and first-request speedups of about 5–7.5× with matching states, logits, tokens and statistics on tested cases (`runs/compiled-runtime`); it is opt-in and is an implementation measurement, not capability. No document score exists. A dated comparator plan (`docs/FRONTIER_COMPARATOR_PLAN.md`) names endpoints `gpt-6-astra` and `claude-fable-5-1` (verified from vendor documentation on 29 September 2026); it matches ordered 72-DPI page rasters and questions, no paid requests have been made and endpoint access is unverified. The measured language failures motivated `docs/LANGUAGE_CORE_OPTIONS.md` (a trained shared trunk, the no-gradient primary, or a three-size study); the owner rejected gradient descent and the trained-core option, so the option is historical. Document scoring is held until a predeclared plain-text QA gate; a proposed 16/20 coherent-answer threshold is not frozen and is not a goal-completion criterion. General language capability, not transport, is the blocking question.

A.7 Evolution pilots on the byte MoE and count-fitted seeds

Generation-zero calibration. `runs/evolution-language-gen0-v2.json` records the founder `f0b43dbe...` on 334 content bytes at 2.042730829833363 bits per byte, zero failures, 9.81954 s single-thread nice-15 (about 34 content bytes/s on this workload). This is selection fitness on public development-training material with EOS excluded; V1's requested 150–256-byte stories selected no cases, and V2 used 288–350 after a 288–600-byte inventory, without using model outputs. A local founder curriculum receipt (`runs/language-curriculum/FOUNDER_DECISION.json`) on eight authored cases scores its 190 content bytes at 4.0663511910499714 bits per byte; raw copy, recall and QA each fail all two cases and all six generations repeat "too" to the 24-token cap (9.64137 s). These cases are separate from the 334-byte calibration, so the losses are not comparable.

Mini-pilot (null). `runs/language-evolution-minipilot-v1-launch.json` records a single-thread nice-15 run with a 2-GiB address-space limit and 600-second timeout: three generations, each proposing one router-array and one expert-array candidate, giving six GA evaluations and six founder-random controls on paired seeds; each proposal mutates eight nonzero coordinates with relative Gaussian scale 0.05 and survives only on strict development improvement. It completes in 152.67957 s with peak RSS 68,136 KiB (about 66.54 MiB), twelve candidate evaluations, two unique report graphs and a sixteen-entry ledger (`runs/evolution-language-minipilot-v1/result.json`). Both arms score 2.042730829833363 selection bits/byte at generations zero and one and 2.0422575408863954 at generations two and three, choosing the same winner `e4d1e5db...`; report-only loss is unchanged at 2.606790482283684 with zero failures. There is no evolutionary advantage or report improvement, report data are public development material, and seven focused tests pass according to the coordinator. Twelve candidates over 152.68 s do not establish a pure candidate rate. Context/state mutations, zero-coordinate activation and crossover are proposed v2 work.

Count-fitted references and seeds. Side research (`notes/CLAUDE_RESEARCH.md`, `scripts/claude_count_reference.py`) reports byte-level order6 Witten-Bell loss 1.3816 on 165 development documents (89,213 bytes), while founder loss 2.5113 uses only 24 documents (12,894 bytes); the scopes are unmatched and establish no matched gap (Table 7). A feature script fitting on development-training documents except the final 150 (used to select mixture weights) reports 1.3909 count-only and 1.2650 with word context and document cache; a side adaptive mixer reports 1.2222. The mixer is a Hedge/exponentiated-gradient-style update, now prohibited and excluded from all primary arms. These are exposed development numbers with non-gradient grid selection, not QA or LLM capability. Server parity of the exact order6 Witten-Bell IR export on the first 24 development documents (12,894 bytes) gives IR and reference both 1.3852251087 bits/byte, maximum position difference about 2.27×10^{-13}; the fitted corpus excludes prior report families, so this is not an exact-scope comparison with 1.3816. Guarded cached scoring on 190 development bytes takes 0.473 s for a router mutant including certification/hash/compile and 0.302 s warm, with bit difference about 1.14×10^{-13}.

Static count-seed four-arm smoke. The bounded static WB6 smoke run (two proposals per search arm) has zero acceptances and unchanged winners; fitted-evolution and fitted-random proposals duplicate at this budget. Eight cached/full qualification cases have maximum bit difference zero (39.986 s). The run takes 25.766 s for 24 batch evaluations (0.931 batch/s); six unique graphs including two baselines correspond to 0.233 unique graph/s end to end. Fitted corpus losses on

Scoring scope	bits/byte	Note
Order6 Witten–Bell, 165 dev docs (89,213 bytes)	1.3816	counts; side reference
Count-only, weights from last 150 train docs	1.3909	non-gradient grid
+ word context and document cache	1.2650	exposed dev feedback
Side adaptive mixer	1.2222	excluded from primary arms
Order6 IR vs reference, first 24 docs (12,894 bytes)	1.3852251087	parity check
Founder, 24 docs (12,894 bytes)	2.5113	unmatched scope
Founder, 334 bytes	2.042730829833363	server calibration
Founder, 190 bytes (curriculum)	4.0663511910499714	different cases

Table 7: Count-fitted references and founder losses. Scopes differ, so rows are not a matched comparison; all data are exposed development material.

batches A/B are 1.0819295/0.7147698 versus founder 2.271318/1.621583; fitted long-context fitness is worse than founder, and a 0.05 residual improves corpus loss slightly but worsens context and is rejected. Twelve ledger entries verify; CPU seconds and peak RSS were not captured (`runs/wb6-four-arm-static-v1/`). This two-proposal null result is not scaling evidence.

A.8 ARC-AGI-2 development: authored cell library, gene coverage and breeding ties

Protocol and exposure. Grid receipts use `arc-grid/1`: SHA256 task-ID hash order, with residue zero modulo five reserved for audit and nonzero for development. The scorer validates grids of 1–30 cells per dimension with colors 0–9, accepts at most two candidates per test input and computes exact task success (every test input correct within two attempts) with a Wilson interval. The exposed development manifest (DEV200) contains 200 public-training tasks with 222 test inputs; it is already development exposure, and relabeling a subset as audit does not make it untouched. Historical receipts state that the residue-zero audit was reserved, but later server access cannot be certified from local metadata. Two attempts must be produced by graph-owned selection, not host-side expert ranking. No official ARC-AGI-2 or tool-use score is reported, and the BFCL V4 agentic category needs a search backend not configured here.

First specialists. The `grid20/1` graph routes among 20 identity/transpose and cyclic-recoloring experts using demonstration reconstruction error; it and its fixed-identity control solve 0/200 (no invalid outputs, Wilson $[0, 0.0188]$). The `grid-associative/1` mutation combines eight dihedral experts with demonstration-inferred color associations in graph state and solves 3/200 (1.5%, Wilson $[0.0051, 0.0432]$) versus 0/200 with inferred recoloring disabled (receipts `runs/grid-associative/DECISION.json`, `runs/grid20/DECISION.json`). Fixed two-gene concatenation measures pair coverage but is not a top-two router.

Coverage and curriculum. The graph-gene operator measures 28 individual genes and three materialized pairs on DEV200; every pair solves 3/200, with no breeding gain. Pair throughput is about 51 tasks/s and 0.0195 CPU s/task, with whole-process high-water RSS 87,384,064 bytes. The dense curriculum has eight families with 200 selection and 100 validation tasks each (2400 distinct original fingerprints): authored selectors solve the five geometry/color families, adding the bbox gene solves crop, and both portfolios fail tiling and local fill. All four graphs solve the same 3/200 native tasks; only two native tasks receive a train-only color-family annotation (both solved), 198 are unclassified (one solved), so family transfer rates are unavailable. The selector is authored, not

learned. Whole-process RSS is 168.3 MiB with 139.33 s measured scoring; the synthetic generators are CC0-1.0/MIT-0 and competition eligibility is unverified. The `grid-gene-evidence-v1` receipt records 300/300 synthetic successes each for authored fill and count genes but 0/200 native tasks each; the count gene has one demonstration-consistent native task whose query fails; portfolios stay 3/200. The largest-four component gene passes 100/100 synthetic but solves 1/200 native with zero fully consistent tasks (about 14 tasks/s, 140 MiB; `runs/largest4-evidence-v1/REPORT.md`). Dense synthetic success has not supplied native transfer.

Separate co-researcher library. A co-researcher's 57-cell host-enumerated depth-two library reports 53/1000 public training solves, 65 demo-consistent tasks and 12 false positives (18.4615% conditional on consistency, 1.2% of all training tasks), and a once-scored public evaluation result of 0/120 with no consistent chains. A later version progresses from 80/1000 training solves (v2) to 95/1000 (v3) with v3 evaluation still 0/120 and 12 false positives reported. These are read-only external receipts on exposed evaluation tasks, not a fresh final evaluation or CellForge graph result. Full training access also potentially exposes the modulo-five audit for a combined library despite the operator's zero audit calls. Source imports and v2 filenames do not pin the receipt-generating run; worker-row means are 1.52686 s training and 1.805416 s evaluation, not a verified 0.5 s CPU budget claim, and RSS is unavailable. Its across-query attempt criterion differs from per-input acceptance, so counts are not a matched comparison. Prize-winning odds are assessed qualitatively as very low; realistic deliverables are a valid open entry and an honest paper.

Single-gene and composition measurements. Table 8 lists authored genes (synthetic denominators are original generated tasks; native means DEV200 unless noted). Each row is one frozen graph with verified ledgers and no gradients, public evaluation, audit or final access. The object-association gain was identified by a post-hoc frozen-model observer, complete paired regression attribution is absent, and the child adds operations without matched compute; the 16-step variant's paired check against the unchanged portfolio verifies one gained task and no losses across 200 tasks, and full pairing against 64 steps is unavailable (aggregate receipt). The two scaling genes' three disjoint successes were already identified from development demonstrations outside DEV200, so this is not new independent transfer. The direction router is an explicit maker-authored correction: a scalar lookup had matched the context graph on all 24 selection tasks with all 97 wrong pixels at unknown keys (94 border pixels, none in conflict with learned keys), and the authored router compares five hypotheses (identity and four one-pixel neighbor copies); authoring cost was not measured. A conditional-state v1 prototype without cells and the zero-cell generic version remain historical.

Training-only diagnostics and coverage census. On DEV200, demonstration-only diagnostics show associative-nine with 2/200 consistent tasks, 57 observed shape mismatches and 141 pixel-rule/routing failures; conditional state has zero consistent, 57 shape and 143 failures (17.014 s, 60.28 MiB, four ledger entries). These categories do not separate missing genes from inadequate routing. A structural census found 57 tasks with shape-changing demonstrations; a bank of identity, square tiling, pixel repetition at factors 2 and 3, and bounding crops explained no task exactly, and one color association added only two tasks the associative portfolio already reconstructed (0.35 s, 38.11 MiB). Extending the 17-rule census to 789 public-training tasks (nonzero SHA256 modulo 5; 211 residue-zero tasks excluded) gave seven exact matches and 13 with color association, two pixel-repetition tasks at factor 2 and one at factor 3 outside DEV200, no object-removal match, and no unique hole-filling match (5.68 s, 53596 KiB). These motivated the scaling genes; they are

Graph / gene	Synthetic	Native	Cost and note
Conditional state v2 (2 rank-cell params, 1002 state dims, 1000 adjacency rules)	300/300 vs 0/300 (updates off)	1/200 vs 0/200; none fully consistent	117 vs 115 tasks/s; 49.3 MiB; 12.03 s per run
Adjacency-then-fill serial child (195 slots, 1010 state)	300/300 vs 0/300 for either part	3/200 for all; 2 consistent, 0 false positives	46.88 vs 52.04 and 50.40 tasks/s; 30.97/27.26/28.17 s; 55.4 MiB; 57 slot nodes, 1650 expanded
Direction router (2 ranking params, 6 state entries, 207 edges)	200/200 selection, 100/100 validation vs 0	0/200; none consistent	142.17 vs 147.06 tasks/s; 55.48 vs 55.64 MiB
Object association child (+11 params)	300/300 vs 0/300	4/200 vs 3/200	58.37 to 9.82 tasks/s; 61.27 to 150.95 MiB
Same, 16-step variant (174 params)	300/300	4/200; 3 consistent, 0 consistent wrong	9.82 to 12.49 tasks/s; 0.10180 to 0.08006 CPU s/task; 150.95 to 109.32 MiB; 1564 vs 3004 edges, 205 vs 589 constants
Pixel scaling factor 2 / 3 (31 params each)	300/300 each	2/789 and 1/789 (expanded training)	403.23 / 393.26 tasks/s; 0.002481 / 0.002544 CPU s/task; 21.76 s; 76040 KiB
Graph-owned scaling selection (64 params)	300/300 vs 100/300 (updates off)	3/789 vs 0/789	182.42 tasks/s; 0.005483 CPU s/task; 47.72 s; 70896 KiB
Expanded active-slot graph (7 slots, 273 params, state 1515)	110/120 vs 70/120 (2 specialists active)	not scored	14.44 vs 44.31 tasks/s; 44.31 s wall; 103064 KiB; 40 paired gains, 0 losses
Homologous five-gene bank recombination (25 combos, 206 params)	child 10/10 vs 5/10 per complementary parent	not scored	14.12 tasks/s; 7.20 s; 113188 KiB

Table 8: Earlier authored ARC genes and compositions on exposed development data. These rows reach 3/200–4/200 on DEV200; subsequent targeted marker additions reach 5/200, as described below. All native gains are exposed development results.

demonstration matches, not query solves. Derived cells must retain expanded-exposure provenance.

Matched population pilots. Table 9 collects every crossover, mutation and random-search comparison. Each is a single seed or a few seeds with four to sixteen proposals on exposed synthetic families, with winners frozen before reporting. Physical evaluation counts match across arms but CPU cost does not. No pilot shows a crossover-specific advantage. Details that change interpretation. In the object-bank pilot paired native receipts confirm one gained task and no losses versus either control, but crossover kept object and adjacency genes while random kept crop and adjacency, so equal totals conceal different niches. In the seed replication all 12 crossover proposals failed to combine object and adjacency, both founders died at generation 3 and two populations were extinct after generation 4; early founder loss is a hypothesis, not an explanation, and pure crossover can only reach alleles in living parents. The age-8 run exceeds the pilot horizon and does not support early founder death as the repair. In the seven-gene pilot all four crossover donor draws were $[0, 0]$ (copying one parent), so the complementary child was never evaluated; crossover and mutation share one winner and one report, so these are not independent. In the separate-streams run distinct winners retained repeat2/repeat3, crop/repeat2 and repeat2/fill, but stream, budget and seed changed together. In the seven-slot run founders scored 14 and 18 on selection, crossover first reached 22 at proposal 10 and mutation at 9, tiling failed in every arm and random also missed object tasks. In the context-projection pilot nine binary key-projection slots were varied with the center retained and all arms kept the orthogonal-cross founder; the graph stores 23 parameter slots. Population mortality and sharing were not separately ablated (shared graphs store 195 slots and 1010 state dimensions), the matched-v1 age/tie rules also changed, and the population-v2 policy lacks a native file-hash manifest, so external byte identity is not asserted.

Nine-slot replication and execution parity. Search seeds 773611, 773621 and 773631 were predeclared with distinct selection and report seeds. Each run uses fourteen authored families, 28 selection tasks and 70 fresh report tasks, two complementary founders, sixteen proposals per arm, carrying capacity four and maximum age eight. Crossover scores 60, 50 and 60; mutation scores 55, 56 and 60; random search scores 65, 66 and 61. Each arm receives eighteen physical evaluations and 1404 predictor calls per run; no evaluation fails and no reported demo-consistent prediction is wrong. All 324 ledger entries, graph identities, corpus commitments and independent random streams were verified. Sparse founders and whole-bank random proposals have different reachable distributions, so this result applies to the declared seeded setting. The genome contains nine typed cell-choice slots in a ten-gene bank (365 stored parameters, 1717 state values); this tests wiring search, not evolution of numerical weights. An opt-in demonstration-state cache preserves exact ranked outputs and candidate trajectories: the paired smoke replay retains all eighteen artifact identities and scores, with 1.56 times the aggregate selection throughput. An alternating frozen runtime benchmark measures 1.95 times the throughput. These are execution improvements, not capability gains.

Staged numerical repair pilot. Three random-stream replications test two rounds of actual weight edits on the already-known horizontal two-marker family. Distinct authored parents are numerically recombined into an OR gate; the maker reads saved training failures, changes a scalar bias, then writes horizontal direction weights. Random repair edits the same tensor from the same discrete value vocabulary. Each arm receives six candidate evaluations and 240 selection predictor calls. Strict error improvement on one batch and nonincrease on a disjoint batch select candidates; other proposals roll back. Frozen fresh synthetic report scores are maker 60/60, random 6/60,

Pilot	Synthetic report (crossover / mutation / random)	Native DEV200 and ledger	Cost
One-slot maker (`grid-maker-slot-v2`, seed 662021)	maker and control pick same fill gene and child; 600/800 to 700/800 (reconstruction 0.75 to 0.875)	3/200; 2400 demo-pair scorings per arm	172.02 s; 75,032 KiB
`grid-population-pilot-v2` (4 logical attempts)	140/180 each	3/200 each; 29 ledger entries; 9 unique genomes; cache hits 3/0/2	150.57 s; 73,804 KiB
`grid-population-matched-v1` (6 physical evals, 3120 calls, 18 evals total)	120/180, 140/180, 140/180	3/200 each; 36 ledger entries	228.63 s; 70,388 KiB
Context-projection (4 founders, 8 proposals, 24 tasks)	1/24 development, 0/24 report for all arms, 95.83% pixel accuracy	0 accepted proposals; 41 ledger entries	0.654–0.684 evals/s; 59.39 s; 123.54 MiB
Object bank (2 founders + 4 proposals, 348 calls per arm, 206 params, 1010 state)	14/20, 12/20, 14/20	4/200, 3/200, 3/200; 36 ledger entries	CPU 18.785, 9.704, 13.099 s; 11.59, 46.94, 47.15 tasks/s; 155.77 s; 140.66 MiB
Seed replication (3 seeds, 20 tasks, 18 evals, 1044 calls)	12/20 every run; mutation 14/20 throughout; random 14, 12, 14	not scored; 36-entry ledgers	CPU 38.675, 54.880, 45.086 s
Finite lifetime (max age 3 to 8, same seeds)	crossover 12/20; mutation 14,14,14 to 14,12,14; random 14,12,14	not scored	18 evals, 1044 calls
Seven-gene (seed 772081; 6 graphs, 420 calls)	12/24, 12/24, 14/24	not scored; 36 ledger entries	CPU 8.171, 8.262, 8.824 s; 0.734, 0.726, 0.680 evals/s; 30.34 s; 114324 KiB
Separate streams, 16 proposals (seed 772101; 18 evals, 1260 calls)	14/24 each (tie)	not scored; 108 ledger entries	CPU 26.575, 39.178, 30.947 s; 105.76 s; 152816 KiB
Seven-slot (seed 772201; 18 graphs, 1260 calls)	22/24, 22/24, 20/24	not scored; 108 ledger entries	CPU 89.416, 139.494, 118.018 s; 11.77, 11.70, 36.92 tasks/s; 372.55 s; 167512 KiB
Nine-slot (3 predeclared seeds; 54 evaluations and 4212 calls per arm)	170/210, 171/210, 192/210	not scored; 324 ledger entries	CPU 117.627, 157.070, 213.305 s; peak RSS 178016256 bytes

Table 9: Breeding, mutation and random-search pilots on the ARC gene banks. Mutation and random figures follow the crossover column order. No comparison establishes a crossover-specific advantage; random search leads in all three nine-slot replications.

unchanged 0/60; random has one demo-consistent wrong prediction. Random beats the maker's intermediate selection errors in two runs. All nine receipt chains and selection decisions were verified; report data were not rescored. Report process peak RSS is 35,524,608 bytes on Windows. The author knows the target rule: this is a numerical feedback-loop pilot, not blinded discovery, independent ARC transfer, maker-parameter learning or archive admission. Authoring-token cost and whole-stage peak memory are unmeasured.

Targeted coverage and capacity. A marker-mirror cell authored from training-only failure diagnostics adds one exposed DEV200 task, raising coverage from 4/200 to 5/200 without paired losses. Joint composition initially displaces an object gene; a three-slot maker repair restores that task and ties the earlier marker portfolio at 5/200. The repair was informed by known diagnostics and is not blind maker efficacy. Expanding seven active slots to nine improves fresh synthetic coverage from 24/28 to 28/28 but leaves native coverage at 5/200 in both arms, with zero false positives and approximately ten percent more query CPU (0.093 versus 0.102 seconds per task). It does not establish independent transfer.

Seven-slot capacity and breeding interfaces. A finite two-slot diagnosis evaluated all 49 layouts (24 selection tasks, 70 demonstrations, 3430 predictor calls; 285.20 s, 190632 KiB, 101 ledger entries): the demonstration-selected winner scored 14/24, post-hoc best single layout 14/24 and the query-oracle union 22/24 (failing both tiling examples); the union is not a deployable score, and one of two scaling examples per factor tests oversized-query identity fallback. These bounds motivated slot/wiring capacity rather than more search in a fixed space. The shared bank then grew to seven authored genes (49 two-slot combinations valid at 268 stored parameters and state 1010; homologous crossover preserves factor 2 and 3 from complementary parents; checks on 12 crossover and 12 mutation children; five tests), and crossover and one-slot uniform mutation operate on the expanded genome (16 reproducible crossover children and 16 mutations, tamper rejection, four children and eight ledger entries); no offspring fitness was measured for these interfaces. A bounded exact 3-by-3 context memory (23 parameters, 3602 state dimensions, four tests) is frozen but unbenchmarked.

A.9 Evolved cellular-automaton negative control

A co-researcher's scratch numpy experiments (not reproduced here; scripts are outside the repository) evolved tiny neural cellular automata with Gaussian mutation and discrete selection on ARC-AGI-2 public-training same-shape tasks, matched by random and identity controls. A 1488-parameter/T10 variant solved 0/40 tasks after 3000 evaluations per task; synthetic swap loss improved without a solve at 6000 evaluations. A 414-parameter/T3 variant fit synthetic training after 8000–14000 evaluations (1200 evaluations/s on a different six-by-six workload, not attributable to CellForge), with held-out success in one of three configurations, then scored 0/40 training and 0/40 test at 8000 evaluations per task with random and identity both zero; lower pixel loss is not task success. A 313-task ten-core run outside the two-core budget was initially reported pending, then corrected: it stopped at 30/313 with zero solves and is not a benchmark. Raw-weight evolution therefore solves no real task at these budgets, consistent with the need for good seeds noted in related work. The published gradient-trained NCA solves 23 of 172 feasible original-ARC tasks [6]; its GPU timing does not establish CPU performance or no-gradient feasibility. On CellForge hardware, a 1488-parameter/T2 graph passes 5/5 synthetic 30-by-30 identity probes at 93.23 adapter calls/s with 38,268,928-byte peak RSS, and a shallow residual graph (`c985a1ed...`, 414 parameters, 22 microcells, three iterations) satisfies the ladder-micro profile (eight combined and three

focused tests) and returns 5/5 at 69.1217 two-phase calls/s with peak RSS 42,491,904 bytes (about 40.5 MiB), single thread and nice 15 (`runs/nca-shallow-v1/server-identity.json`). These are demo/query adapter calls, not ARC fitness; architecture, iteration count and dispatch differ, so parameter count does not explain the speed gap, and no ARC score, evolution or generalization is established.

A.10 Offline submission interface

An offline CPU submission adapter and source bundler have eight and three focused tests; bundling includes transitive dependencies and hash-manifest preflight. A server smoke test on two tasks and three queries takes 0.14686 s at 48,603,136-byte RSS and matches local JSON exactly (`runs/arc-cpu-package/README.md`). This establishes interface execution, not Kaggle eligibility or task accuracy.

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